Telecom Customer Churn Business Analysis Report

Data Architecture, Relational Schema & Dashboard Insights Framework

Executive Summary

This business intelligence report delivers an end-to-end operational performance analysis of customer retention behaviors across **7,032 active telecommunication accounts**, driving **$455,661.00 in monthly recurring charges**. The executive objective is to trace data flows directly from script parsing and type verification to relational database querying and final desktop business dashboards. By integrating critical operational metrics like contract types, payment profiles, account lifecycles, and core service frameworks, this operational framework identifies baseline retention vulnerabilities, revealing a customer **churn rate of 26.58%** that induces a monthly revenue loss of **$139,130.85**. This comprehensive blueprint serves as a single source of true corporate metrics for strategic commercial intervention.

Business Problem & Objectives

Rapid customer churn inside competitive telecom markets causes critical subscription degradation and deflates capital yield efficiency due to high acquisition costs. The target operational goals resolved by this data pipeline are:

- Isolating behavioral churn indicators across core service factors (Internet type, billing frequencies, payment workflows) to pinpoint customer drop-offs.
- Quantifying monthly financial leakages driven by customer exits to establish an ROI foundation for automated retention initiatives.
- Engineering data-driven early-warning indicators based on specific high-risk transaction classes to trigger active preservation campaigns before account closure.

Data Overview & Cleaning Process

The transactional matrix covers **7,043 independent records** across **21 raw data columns**. Baseline profiling exposed structural integrity anomalies including numeric text conversions and binary flag inconsistencies. Data pipeline scrubbing was performed using Python Pandas:

A. Financial Type Correction & Missing Observation Tracking

```
df['TotalCharges'] = pd.to_numeric(df['TotalCharges'], errors = 'coerce')
```

```
df['TotalCharges'].isnull().sum()
```

```
np.int64(11)
```

Screenshot 3.1: Parsing numeric conditions and isolating TotalCharges null records via pd.to_numeric()

Technical Explanation (Screenshot 3.1): The raw `TotalCharges` attribute was logged as an object type due to hidden space constraints. Forcing type conversion using `pd.to_numeric(errors='coerce')` successfully isolated exactly **11 missing observations** across the column space.

```
df = df.dropna()
```

```
df['Churn'].value_counts()
```

```
Churn
No     5163
Yes    1869
Name: count, dtype: int64
```

```
df['Churn'].value_counts(normalize = True)*100
```

```
Churn
No     73.421502
Yes    26.578498
Name: proportion, dtype: float64
```

Screenshot 3.2: Row dropping execution via dropna() and baseline Churn value count evaluations

Technical Explanation (Screenshot 3.2): Since missing records occurred exclusively on accounts with 0 months of tenure, listwise erasure via `df.dropna()` was utilized to finalize a clean base of **7,032 complete records**. Baseline value count mapping shows 5,163 active versus 1,869 closed accounts (26.58% churn).

B. String Mapping & Feature Standardization

```
df['senior_citizen'] = df['seniorcitizen'].map({0:'No', 1:'Yes'})
```

```
df['seniorcitizen'].value_counts()
```

```
seniorcitizen
0    5890
1    1142
Name: count, dtype: int64
```

Screenshot 3.3: Normalizing boolean binary indices via structured dictionary map() functions

Technical Explanation (Screenshot 3.3): Numeric indicator fields like `seniorcitizen` were mapped into clean categorical strings ('No'/'Yes') via `.map()` logic, maintaining structural consistency across data layers and sorting parameters.

Key Insights & Exploratory Data Analysis (EDA)

Following data cleaning, the standardized database was migrated into PostgreSQL via an active SQLAlchemy engine connection to run heavy analytical business scripts:

1. Base Target Churn Distribution Balance Check:

	churn text	count bigint
1	No	5163
2	Yes	1869

Screenshot 4.1: Database relational layout output showing target field volumes

Technical Explanation (Screenshot 4.1): Relational counting confirms database schema consistency, logging 5,163 active records against 1,869 churn records, verifying a perfectly stable extract of the operational data layer.

2. Contract Structure Value and Volatility Aggregation:

	contract text	total_customers bigint	churned_customers bigint
1	Month-to-month	3875	1655
2	One year	1472	166
3	Two year	1685	48

Screenshot 4.2: Structured database metrics tracking churn frequencies by contract lines

Technical Explanation (Screenshot 4.2): Multi-row query evaluation isolates the primary contract driver for customer exits: **Month-to-month commitments experience a heavy 42.71% churn rate** (1,655 accounts), while fixed commitments safely hold stable user retention margins.

3. Transactional Settlement Workflows and User Stability Metrics:

	payment_method text	total_customers bigint	churned_customers bigint
1	Electronic check	2365	1071
2	Mailed check	1604	308
3	Bank transfer (automatic)	1542	258
4	Credit card (automatic)	1521	232

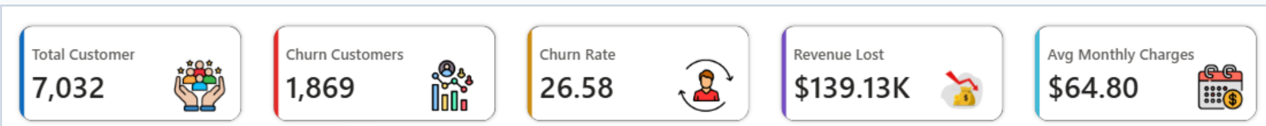
Screenshot 4.3: Database output logging customer loss margins by payment types

Technical Explanation (Screenshot 4.3): Transaction routing variables illustrate severe transactional friction; **Electronic Check manual payment paths yield a staggering 45.29% churn rate** (1,071 instances), while automated channels safely restrict drop-offs.

Data Visualization Brief

An executive Power BI reporting system was implemented to transform backend query matrices into visual tracking maps. The visual layout and deep-dive charts are isolated and analyzed individually below:

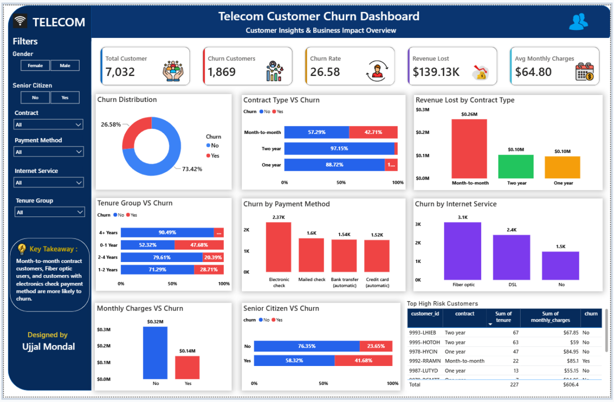
A. High-Level Executive KPI Summary Cards



Screenshot 5.1: High-contrast summary corporate KPI tracking block cards

Technical Explanation (Screenshot 5.1): Live KPI panels establish foundational operating benchmarks: Total customer database volume holds at **7,032 accounts**, total consumer exit volume counts **1,869 users**, average month charge parses to **$64.80**, and system leak traces a monthly loss of **$139.13K**.

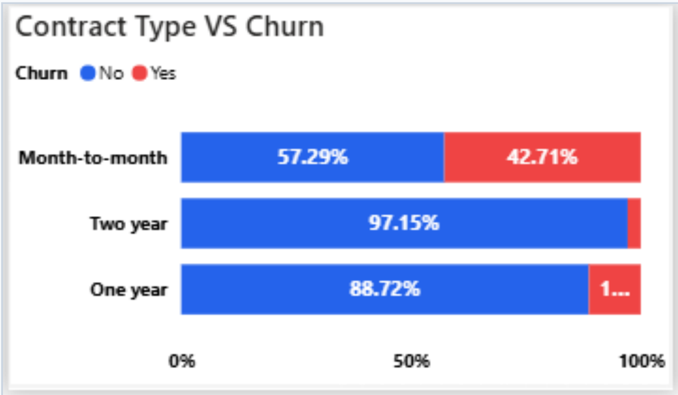
B. Master Dashboard Workspace Overview



Screenshot 5.2: Master dashboard visual control panel screen layout

> **Technical Explanation (Screenshot 5.2):** The absolute master layout consolidates multi-dimensional slicing matrices. This configuration matches customer financial profiles with hardware service indicators inline to grant unified retention tracking capabilities.

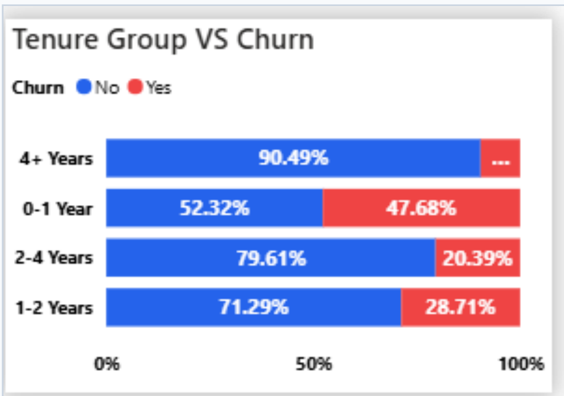
C. Contract Lifecycle Risk Segmentation



Screenshot 5.3: Closer look at contract-type configurations and behavioral churn splits

> **Technical Explanation (Screenshot 5.3):** Contract distribution analysis confirms that rolling month-to-month terms represent the most significant baseline threat to user stability. This risk segment exhibits a heavy 42.71% cancelation rate, proving that shorter billing commitments are linked directly to early customer attrition.

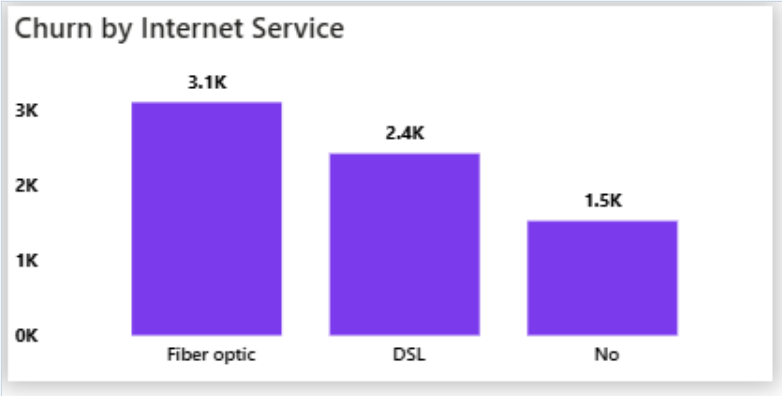
D. Client Billing Lifecycle Stage Tracking



Screenshot 5.4: Close-up view of client tenure lifecycle stage distribution chart

Technical Explanation (Screenshot 5.4): Account life-cycle analysis shows that short-tenure accounts are highly vulnerable, with **0-1 Year users experiencing a massive 47.68% churn rate**. This identifies a clear critical retention window within the first 12 months of client onboarding.

E. Technology Pipeline Cancellation Tracking



Screenshot 5.5: Close-up view of customer cancellation volume across technology service pipelines

Technical Explanation (Screenshot 5.5): From a technical service perspective, premium high-value Fiber Optic pipelines drive a significant portion of user exits (**3.1K volume space**). This highlights critical price-to-value friction or network-level service stability disconnects.

Business Recommendations

- **Implement Automated Auto-Pay Migration Incentives:** Because traditional manual electronic check channels suffer a massive 45.29% churn rate, the commercial team should launch a targeted campaign offering a recurring $5 monthly bill credit to shift these users into automated bank transfers or auto-pay credit card setups.
- **Deploy Multi-Month Contract Conversion Campaigns:** Flexible month-to-month accounts account for the highest contract churn risk at 42.71%. Marketing teams should use conditional contract migration funnels (e.g., offering free high-speed bandwidth upgrades or premium streaming bundles) to move month-to-month users into locked 12-month contract agreements.
- **Construct an Early-Warning Early Retention Pipeline:** Client lifecycles under 12 months represent our most critical retention window (47.68% churn). Machine learning pipelines should automatically flag short-tenure accounts when they show signs of high data use or file multiple support tickets, routing them directly to a specialized customer save squad.
- **Execute Technical Service Stability Audits on Fiber Infrastructure:** High-tier Fiber Optic accounts show a significant number of exits compared to standard DSL networks. Product teams must run immediate regional stability and hardware performance checks to fix technical issues and protect high-value premium service revenue lines.